

supply chain aspect, Jhanwar says the reconfiguration is for greater reliability and resilience.

For their part, firms think that the government's initiatives such as Make in India and the push for electrification augur well for the country. "We launched our first luxury electric vehicle (EV) in India, which is running successfully," claims Kulkarni. Over time, the confidence levels of those putting in the money have taken off too. "It is not just OEMs or product makers who are evolving, but India has become a hub for many global Tier I suppliers as well. A mature ecosystem with skilled manpower, cost advantage and digital prowess makes us believe that we can enhance our manufacturing footprint in India," he says. Speaking of sustainability, his company has set a target of making its new fleet of passenger cars CO₂-neutral over the entire life cycle of the vehicle by 2039.

An important aspect of manufacturing is to find ways to grow exports. In defence, for instance, domestic production is around ₹80,000 crore, with the government wanting to increase that to ₹1.75 lakh crore by 2025. Compare that to exports in 2014, which amounted to ₹500 crore, and is now at ₹15,000 crore. "By itself, that

resultant progress would be multifold. "Creating an ecosystem means suppliers are locked in, making the whole process of doing business extremely conducive."

Since 2047 is still a quarter of a century away, there will be many things that will come to fruition along the way. India in the next two decades, Jhanwar predicts, will be a hi-tech and R&D-intensive manufacturing economy. "An extensive adoption of digitalisation, robotics and analytics would provide a multiplier effect or the key 'productivity to prosperity' driver," he says. To him, the multitude of factories and machinery generating copious amounts of data is the foundation to harnessing deeper insights. "In fact, the cascading effects of policy, tech and research-based growth will lead to many SMEs making the transition to manufacturing unicorns. Technology for the future will primarily originate from the start-up ecosystem."

Think manufacturing and go global is the credo. "We are manufacturing for the world. India is looking to exploit its digital strengths, take that to manufacturing and provide a robust alternative to China," explains Ramchandani. He says there is little doubt that by

102 |

THERE IS OPTIMISM IN THE MANUFACTURING SECTOR AS IT IS FLUSH WITH A TALENTED WORKFORCE AND BACKED BY INVESTMENTS FROM LARGE GLOBAL AND DOMESTIC FIRMS

is not a large number and we are a small player with a presence across segments. A defence industrial complex is being created," says Ramchandani. That said, he expects India to be a larger player in 2047, and defence to account for "a healthy share of manufacturing".

In the midst of all this, the government has a role to play too. ABB's Sharma attempts to highlight a difference between manufacturing and everything else. "If you look at services, the progress has been tremendous, but it was driven by IT infrastructure. That is now at a high stage of maturity and India is well placed," he explains. However, manufacturing calls for physical infrastructure and that is not something that can be glossed over. Citing cases of nations such as Mexico, Vietnam, Malaysia and, of course China, he elaborates on how the infrastructure was created much before industry came in. "That [infrastructure] could be the existence of vast tracts of land or high-quality ports. It gives industry high levels of comfort when all these are in place," says Sharma. India will need to adapt to this template, and if that does take place, the

2047, India will have made the transition from relatively small-scale manufacturing to becoming a global manufacturing hub. Digital alone has the ability to ramp up the manufacturing sector. Enabling free-flowing access to information on the digital backbone helps in paving the way for higher productivity and transfer of knowledge. If Industry 4.0 is here, maybe, we could be seeing an Industry 7.0 in 2047. That's how much of a profound impact digital can have, Ramchandani explains.

Looking into the future, 'manufacturing next' is a phrase Jhanwar picks to define an industry resilient to frequent disruptions. "It could be supply chain shocks, technology shifts, volatile energy prices, product obsolescence or innovative business models. India has access to all the key ingredients and a great foundation to become a leading player in manufacturing next," he says, with a generous dose of hope and confidence. Our place under the manufacturing sun is just around the corner. **BT**

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